LABORATORY · Lab · 26 hours

Experiment 4: Characteristics of Synchro Transmitter and Receiver Pair



Control Systems Laboratory (22EEG53)

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Learning objectives

- Study the construction and principle of a synchro pair
- Plot the receiver rotor angle versus transmitter angle
- Plot the error voltage versus angular displacement
- Understand its use as an error detector

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and apparatus

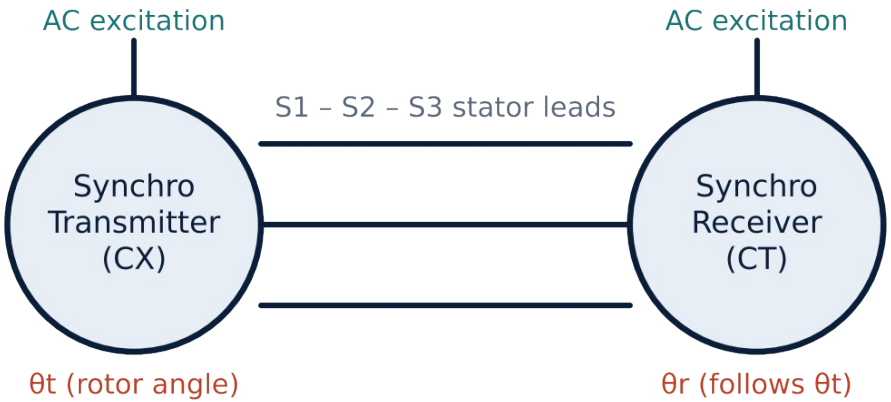
Aim: to study the characteristics of a synchro transmitter and receiver pair.

Apparatus: synchro transmitter (CX) and control transformer/receiver (CT) trainer kit, AC supply, angular scales, voltmeter, connecting leads.

Use: error detector in position control systems.

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Synchro pair



Connection of synchro pair

- Transmitter rotor excited from AC supply.
- Three stator windings S1, S2, S3 at 120°.
- Stator leads interconnected to the receiver.
- Receiver output ∝ sin(θt – θr).

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Theory

Synchro transmitter (CX)

Single-phase rotor, three-phase stator. Stator voltages ∝ V cos(θ − 120°k).

Control transformer (CT)

Rotor output voltage ∝ sin(θt − θr): the error signal, used with an amplifier.

Error detector

For small error e ≈ θt − θr: Ve = Ks (θt − θr), Ks in V/rad.

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Procedure

1

Connections

Connect S1–S1, S2–S2, S3–S3 of CX and CT; excite rotors from the same AC supply (as per kit).

2

Set zero

Align both pointers to 0°.

3

Vary θt

Rotate the transmitter in steps of 10° (0°–360°).

4

Record

Note receiver angle θr or error voltage Ve for each step; plot graphs.

Safety: keep the supply off while making connections.

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Observation table

θt (deg)	θr (deg)	Error θt − θr	Error voltage Ve (V)
0			
30			
60			
90			
120			
…	…	…	…

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Expected results

- θr follows θt closely: straight line at 45° (θr = θt).
- Error voltage is sinusoidal in the error angle: zero at 0° and 180°, maximum at 90°.
- Near zero error the relation is nearly linear: sensitivity Ks.
- Small residual error arises from friction and loading.

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Key takeaways

- Synchro pair transmits angular position electrically
- CT output ∝ sin(θt − θr): error signal
- Linear near the null; used as error detector
- Plot θr vs θt and Ve vs error angle

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Practice questions

Q1. Explain the construction of a synchro transmitter.

Q2. Why is the CT output zero when the rotors are aligned?

Q3. Define synchro sensitivity.

Q4. Name two applications of synchros.